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Education

KAIST

PH.D., SCHOOL OF COMPUTING

Daejeon

2023.02 - present

- Advisor: Professor Joyce Jiyoung Whang

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M.S., SCHOOL OF COMPUTING

Daejeon

2021.08 - 2023.02

- Advisor: Professor Joyce Jiyoung Whang

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B.S., SCHOOL OF COMPUTING

Daejeon

2017.02 - 2021.08

- Double Major: Department of Mathematical Sciences

Publications

† equal contribution, * corresponding author

1. **Jaejun Lee** and Joyce Jiyoung Whang*, “Learn When and Where to Connect: Adaptive Virtual Nodes for Dynamic Message Passing on Graphs”, *Proceedings of the 32nd SIGKDD Conference on Knowledge Discovery and Data Mining (KDD)*. To be published.
2. **Jaejun Lee**, Seheon Kim, and Joyce Jiyoung Whang*, “Generative Representation Learning on Hyper-relational Knowledge Graphs via Masked Discrete Diffusion”, *Proceedings of the 43rd International Conference on Machine Learning (ICML)*. To be published.
3. **Jaejun Lee** and Joyce Jiyoung Whang*, “Structure Is All You Need: Structural Representation Learning on Hyper-Relational Knowledge Graphs”, *Proceedings of the 42nd International Conference on Machine Learning (ICML)*, pages 33765-33783, July 2025.
4. Minsung Hwang, **Jaejun Lee**, and Joyce Jiyoung Whang*, “Stability and Generalization Capability of Subgraph Reasoning Models for Inductive Knowledge Graph Completion”, *Proceedings of the 42nd International Conference on Machine Learning (ICML)*, pages 26376-26411, July 2025.
5. **Jaejun Lee**, Minsung Hwang, and Joyce Jiyoung Whang*, “PAC-Bayesian Generalization Bounds for Knowledge Graph Representation Learning”, *Proceedings of the 41st International Conference on Machine Learning (ICML)*, pages 26589-26620, July 2024.
6. **Jaejun Lee**, Chanyoung Chung, Hochang Lee, Sungho Jo, and Joyce Jiyoung Whang*, “VISTA: Visual-Textual Knowledge Graph Representation Learning”, *Findings of the Association for Computational Linguistics: EMNLP 2023 (Findings of EMNLP)*, pages 7314-7328, December 2023.
7. Chanyoung Chung†, **Jaejun Lee**†, and Joyce Jiyoung Whang*, “Representation Learning on Hyper-Relational and Numeric Knowledge Graphs with Transformers”, *Proceedings of the 29th ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD)*, pages 310-322, August 2023.
8. **Jaejun Lee**, Chanyoung Chung, and Joyce Jiyoung Whang*, “InGram: Inductive Knowledge Graph Embedding via Relation Graphs”, *Proceedings of the 40th International Conference on Machine Learning (ICML)*, pages 18796-18809, July 2023.
9. Ji Ho Kwak†, **Jaejun Lee**†, Sungho Jo*, and Joyce Jiyoung Whang*, “Semantic Grasping via a Knowledge Graph of Robotic Manipulation: A Graph Representation Learning Approach”, *IEEE Robotics and Automation Letters*, Vol. 7, No. 4, pages 9397-9404, October 2022. Also presented at 2022 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)

Teaching Experience _____

TUTORIALS

- Assistant Tutor of “Knowledge Discovery with Knowledge Graphs: From Structural Embeddings to Generative Reasoning(KnowKG)”, To Appear in *SIGKDD Conference on Knowledge Discovery and Data Mining(KDD)*, Tutorial, August 2026.
- Assistant Tutor of “Key Facets in Modern Knowledge Graph Representation Learning (KeyKGRL)”, *International Semantic Web Conference (ISWC)*, Tutorial, November 2025.

TEACHING ASSISTANT

CS471 Graph Machine Learning and Mining	Spring 2023, 2024 and 2025	KAIST
CS376 Machine Learning	Fall 2021, 2022, 2023, 2024 and 2025	KAIST
CS665(DS532) Advanced Data Mining	Fall 2023	KAIST
CS492 Special Topics in Computer Science: Graph Machine Learning and Mining	Spring 2022	KAIST

Professional Services _____

PEER REVIEW

- NeurIPS 2025 Reviewer
- ICML 2026 Reviewer - Selected as a Gold Reviewer
- KDD 2026 Reviewer
- ACL ARR Reviewer
- TKDD Reviewer
- TMLR Reviewer